Estimating Our First Parameters

ES 335 / ES 678 Machine Learning, Sem I 2026 Source: `sources/course-2026/lectures/L03_student_estimating_parameters.pdf` (29 slides); board annotations in `L03a_…` / `L03b_…` · Previous: Mathematical Foundations

How to use this file. Part A builds the *new* maths Lecture 3 needs, on top of Lectures 1–2 (probability, densities, the normal, logs, derivatives, argmin/argmax, Π and likelihood). Part B goes through all 29 slides. The "In Class / Work It Out" steps are left blank in the PDF, so they are **fully solved** here. Part C is practice.

What this lecture is really about: one recipe, used for the rest of the course:

$$\underbrace{p(x_i \mid \theta)}_{\text{one reading}} \xrightarrow{\prod_i} \underbrace{L(\theta)}_{\text{independence}} \xrightarrow{\log} \underbrace{\ell(\theta) = \sum_i \log p(x_i \mid \theta)}_{\text{logs}} \xrightarrow{\frac{d}{d\theta}=0} \underbrace{\hat{\theta}}_{\text{the estimate}}$$

The recipe works the same whatever distribution you pick. When data are scarce, add a **prior** and you get **MAP**, plus a measure of **how uncertain** θ is.

Step-up path

You need. Derivatives, logs, and the idea of a probability distribution.

1. Fix the data, and let the parameter vary: that is a likelihood.
2. Take logs, because products become sums and the peak stays in place.
3. Differentiate, set to zero, and solve: that is the MLE recipe.
4. Add a prior to get a posterior, and read the MAP and posterior mean off it.
5. Ask how sure you are, by looking at how sharp the peak is.

You can stop when you can derive the MLE of a new distribution without looking anything up.

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Part A Maths From the Ground Up

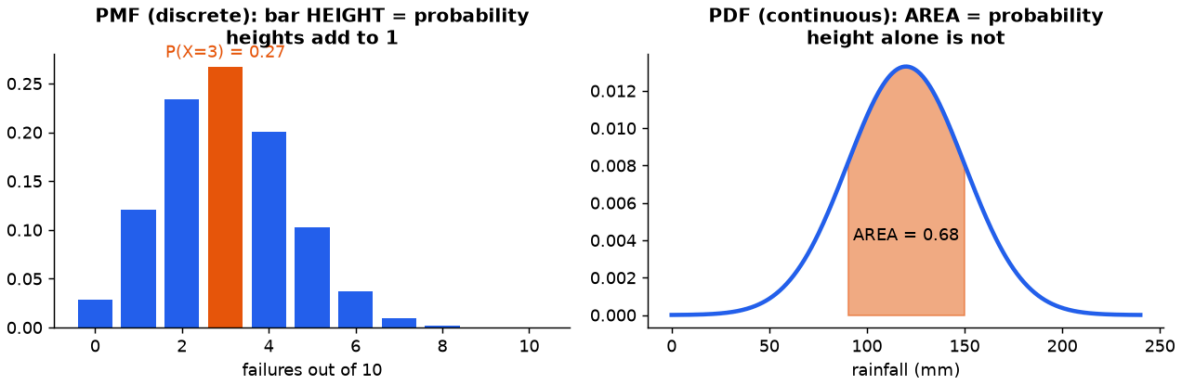
A1 Data vs parameters, estimators and hats

From now on every problem has two kinds of symbol:

Symbol	What it is	Who provides it	Lecture 3 example
$x = (x_1, \ldots, x_n)$	**data**	handed to you	40 recorded flood-day rainfalls
θ	**parameters**, the numbers inside the model	unknown, so **you guess them**	centre μ and spread σ
$\hat{\theta}$	**estimate**, your guess	computed from x	$\hat{\mu} \approx 118$ mm, $\hat{\sigma} \approx 31$ mm

- An **estimator** is a *rule* that turns data into a guess: $\hat{\theta} = g(x_1, \ldots, x_n)$. Example: "add them up and divide by n".
- An **estimate** is the number that rule gives on *this* dataset.
- The **hat** (L1 §2) means "computed from data". θ is the true unknown value; $\hat{\theta}$ is our guess. Different data would give a different $\hat{\theta}$, so an estimator is itself a random variable (this matters in L8: bias and variance of estimators).

A2 Distributions: PMF vs PDF



Discrete: read the bar height. Continuous: read an area, never a single height.

A **distribution** is a rule that says how often each value shows up. It has three parts:

- **support**: which values are allowed (rainfall ≥ 0; a coin $\in \{0, 1\}$);
- **parameters**: the adjustable numbers inside the rule (μ, σ; θ; λ);
- **normalisation**: the total must be 1.

Discrete → probability mass function (PMF).

$$p(x) = \Pr(X = x), \qquad \sum_x p(x) = 1, \qquad 0 \leq p(x) \leq 1.$$

Each value carries an actual probability.

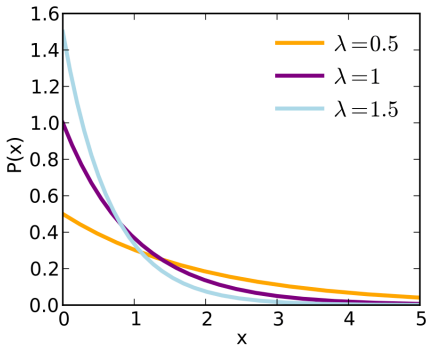
Continuous → probability density function (PDF).

$$\Pr(a \leq X \leq b) = \int_a^b p(x)\,dx, \qquad \int_{-\infty}^{\infty} p(x)\,dx = 1, \qquad p(x) \geq 0.$$

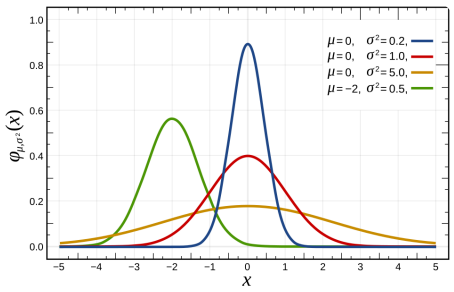
Careful. **A density is not a probability. Only its area is.** For a continuous X, $\Pr(X = 90) = 0$ exactly (a line has zero area). A density can also be **bigger than 1**: Uniform on $[0, 0.5]$ has $p(x) = 2$ on that interval, and the area is still $2 \times 0.5 = 1$.

Why use densities in likelihoods, then? Because $\Pr(x \leq X \leq x + \Delta) \approx p(x)\,\Delta$ for a tiny width Δ. Comparing two θ's, the Δ is the same for both, so the density does the comparison by itself.

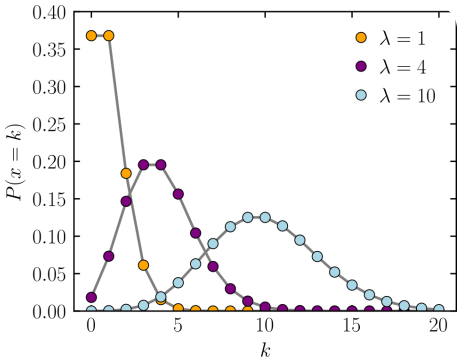
A3 The distribution zoo, with formulas



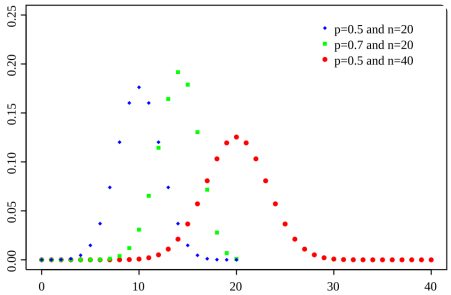
Exponential PDFs: waiting times, highest at 0. (Wikimedia Commons: Exponential_pdf.svg, CC BY 3.0)



Normal PDFs: μ shifts the bell, σ widens it. (Wikimedia Commons: Normal_Distribution_PDF.svg, Public domain)



Poisson PMF for several rates λ. (Wikimedia Commons: Poisson_pmf.svg, CC BY 3.0)



Binomial PMF: bars, because counts are discrete. (Wikimedia Commons: Binomial_distribution_pmf.svg, Public domain)

Discrete (PMF):

Name	Story	PMF $p(x \mid \theta)$	Mean	Variance
Bernoulli(θ)	one yes/no	$\theta^x(1-\theta)^{1-x},\ x \in \{0,1\}$	θ	$\theta(1-\theta)$
Binomial(n, θ)	number of yeses in n tries	$\binom{n}{x}\theta^x(1-\theta)^{n-x}$	$n\theta$	$n\theta(1-\theta)$
Categorical($\theta_1..\theta_K$)	one of K labels	θ_x, with $\sum_k \theta_k = 1$	—	—
Poisson(λ)	counts per interval	$\frac{\lambda^x e^{-\lambda}}{x!},\ x = 0, 1, 2, \ldots$	λ	λ

Continuous (PDF):

Name	Story	PDF $p(x \mid \theta)$	Mean	Variance
Normal(μ, σ^2)	measurements spread around a centre	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$	μ	σ^2
Exponential(λ)	waiting time	$\lambda e^{-\lambda x},\ x \geq 0$	$1/\lambda$	$1/\lambda^2$
Uniform(a, b)	no reason to prefer any value	$\frac{1}{b-a}$ on $[a, b]$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Beta(a, b)	a proportion in $[0, 1]$	$\propto \theta^{a-1}(1-\theta)^{b-1}$	$\frac{a}{a+b}$	see A9

The Bernoulli trick: $\theta^x(1-\theta)^{1-x}$ is θ when $x = 1$ and $1-\theta$ when $x = 0$. It's one formula covering both cases, which makes products easy (slide 22).

Notation: $x!$ ("x factorial") $= x(x-1)\cdots 1$, with $0! = 1$. $\binom{n}{x} = \frac{n!}{x!(n-x)!}$ counts the ways to choose which x of the n tries were yeses.

A4 Why these shapes keep appearing

These shapes aren't arbitrary conventions. Each one follows from an assumption about the **mechanism** that produced the data:

If you believe...	...you are forced into	Why (one line)
many small independent effects add up	**Normal**	Central Limit Theorem: sums of many small independent pieces look bell-shaped
rare events, many chances to happen	**Poisson**	Binomial with n huge, θ tiny, $n\theta = \lambda$ fixed → Poisson
waiting with no memory of time already waited	**Exponential**	the only continuous distribution with $\Pr(T > s + t \mid T > s) = \Pr(T > t)$
one trial, two outcomes	**Bernoulli**	nothing else is possible

The choice also decides your **loss function**. A normal assumption leads to squared error $(y - \hat{y})^2$ (Lecture 4, and L2 slides 32–36). A Bernoulli assumption leads to cross-entropy (logistic regression, L6).

A5 Likelihood is not probability

The same formula $p(x \mid \theta)$ can be read two ways:

Reading	Fixed	Varies	Name	Sums/integrates to 1?
forwards	θ	x	probability / density of data	**yes**, over x
backwards	x (already happened)	θ	**likelihood** $L(\theta)$	**no**, not over θ

Example: Bernoulli, one observation $x = 1$. Then $L(\theta) = \theta$ on $[0, 1]$, and $\int_0^1 \theta \, d\theta = \frac{1}{2} \neq 1$. So the likelihood is not a distribution over θ. It's a **score**: higher means "this θ makes my data less surprising".

For independent data the scores multiply (L2 §A14):

$$L(\theta) = \prod_{i=1}^{n} p(x_i \mid \theta).$$

A6 Logs: products to sums, peaks stay put

The rules you need (natural log, $\log = \ln$):

Rule	Example
$\log(ab) = \log a + \log b$	$\log \prod_i p_i = \sum_i \log p_i$
$\log(a^k) = k \log a$	$\log \theta^k = k \log \theta$
$\log(1/a) = -\log a$	$\log \frac{1}{\sqrt{2\pi\sigma^2}} = -\frac{1}{2}\log(2\pi\sigma^2)$
$\log e^u = u$	$\log e^{-(x-\mu)^2/2\sigma^2} = -\frac{(x-\mu)^2}{2\sigma^2}$
$\frac{d}{d\theta}\log\theta = \frac{1}{\theta}$	
$\frac{d}{d\theta}\log(1-\theta) = -\frac{1}{1-\theta}$	chain rule: inner derivative is -1

Why logs are allowed: $\log$ is strictly increasing, so if $L(\theta_1) > L(\theta_2)$ then $\log L(\theta_1) > \log L(\theta_2)$. The argmax doesn't move (L2 §A8):

$$\arg\max_\theta L(\theta) = \arg\max_\theta \log L(\theta).$$

Why logs are needed: 40 densities around 0.01 multiply to about 10^{-80}, which is close to the limit where a computer rounds to 0 (*underflow*). Its log is $-80 \ln 10 \approx -184$, an ordinary number.

A7 Finding a maximum: derivative = 0, then check

For a smooth function of one variable:

1. Differentiate: $\ell'(\theta)$.
2. Set $\ell'(\theta) = 0$ and solve. These are the *stationary points*.
3. Check that it's a **maximum**: $\ell''(\theta) < 0$ (curving down, like a hill). If ℓ is concave everywhere, the stationary point is *the* global max.
4. Check the edges of the allowed range (e.g. $\theta = 0$ or 1). With little data the peak can sit **on the edge** (slide 25).

Several parameters (μ and σ): take each **partial derivative** (L2 §A6), set *all* of them to 0, and solve together. In the normal case they decouple: $\hat{\mu}$ doesn't depend on σ, so you solve for μ first, then plug $\hat{\mu}$ into the σ equation.

Useful fact: the constant terms (those without θ) disappear when you differentiate. That's why we can ignore $-\frac{n}{2}\log(2\pi\sigma^2)$ when solving for μ.

A8 Bayes' rule for parameters: prior, likelihood, posterior

L1 §8.8 used Bayes' rule on events. Use it on θ itself (this was written on the board):

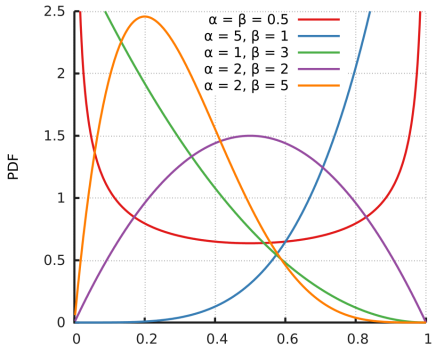
$$\underbrace{p(\theta \mid x)}_{\text{posterior}} = \frac{\overbrace{p(x \mid \theta)}^{\text{likelihood}}\ \overbrace{p(\theta)}^{\text{prior}}}{\underbrace{p(x)}_{\text{evidence}}}, \qquad p(x) = \int p(x \mid \theta)\, p(\theta)\, d\theta.$$

- **Prior** $p(\theta)$: what you believed about θ *before* the data.
- **Likelihood**: what the data say (A5).
- **Posterior**: what you believe *after* the data. Unlike the likelihood, it is a real distribution over θ.
- **Evidence** $p(x)$: the marginal probability of the data. It doesn't depend on θ, so for finding the peak we only need

$$p(\theta \mid x) \propto p(x \mid \theta)\, p(\theta).$$

($\propto$ = "proportional to": equal up to a constant that doesn't involve θ.)

A9 The Beta distribution and conjugacy



Beta(a, b) shapes: flat, hill, U, skewed. All live on [0, 1]. (Wikimedia Commons: Beta_distribution_pdf.svg, Public domain)

For a proportion $\theta \in [0, 1]$:

$$\text{Beta}(\theta \mid a, b) \propto \theta^{a-1}(1-\theta)^{b-1}, \qquad a, b > 0.$$

Quantity	Formula	Beta(2,2)
mean	$\bar{\theta} = \dfrac{a}{a+b}$	0.5
mode (peak), for $a, b > 1$	$\dfrac{a-1}{a+b-2}$	0.5
variance	$\dfrac{\bar{\theta}(1-\bar{\theta})}{a+b+1}$	$0.25/5 = 0.05$

Shapes: Beta(1,1) is flat (uniform). Beta(2,2) is a gentle hill at 0.5 ("probably not 0 or 1"). Large a, b give a sharp spike.

Conjugacy. Multiply a Beta prior by a Bernoulli/Binomial likelihood $\theta^k(1-\theta)^{n-k}$. The exponents add, and you get a Beta again:

$$\theta^{a-1}(1-\theta)^{b-1} \cdot \theta^k(1-\theta)^{n-k} = \theta^{(a+k)-1}(1-\theta)^{(b+n-k)-1} \;\Rightarrow\; \text{Beta}(a+k,\; b+n-k).$$

A prior whose posterior has the same family is called **conjugate**. Updating is just bookkeeping: add the failures to a and the passes to b.

Pseudo-counts. $a-1$ acts like $a-1$ imagined failures and $b-1$ like imagined passes, seen before the real data.

A10 MLE vs MAP vs posterior mean

Estimate	Definition	Uses prior?	Bernoulli answer (Beta(a, b) prior)
MLE	$\arg\max_\theta p(x \mid \theta)$	no	$\dfrac{k}{n}$
MAP (maximum a posteriori)	$\arg\max_\theta p(\theta \mid x)$ = posterior mode	yes	$\dfrac{k+a-1}{n+a+b-2}$
Posterior mean	$\mathbb{E}[\theta \mid x]$	yes	$\dfrac{k+a}{n+a+b}$

- With a flat prior Beta(1,1), MAP = MLE.
- As $n \to \infty$ the data swamp the pseudo-counts and all three agree.

- MAP in log form: $\hat{\theta}_{\text{MAP}} = \arg\max_\theta \left[\ell(\theta) + \log p(\theta)\right]$, i.e. log-likelihood + a penalty. **This is exactly where regularisation comes from** (L7: ridge = Gaussian prior, lasso = Laplace prior).
- Neither MLE nor MAP gives the uncertainty. The **posterior width** (its standard deviation) does.

Part B Slide by Slide (all 29 pages)

Slide 1 Title

Machine Learning, Lecture 3: Estimating Our First Parameters.

Slide 2 Course Roadmap: You Are Here

L1 ML in 2026 → T1/A0 readiness → L2 learning problem → **L3–L6 estimation and linear models (you are here)** → L7–L8 trust and evaluation → L9–L15 fit models → L16–L17 metrics and validation → L18–L24 model families → L25–L28 modern systems.

Data come from a distribution whose numbers we don't know. **Maximum likelihood** is how we guess them, and the recipe doesn't care which distribution it is.

Slide 3 Where We Got To Last Time

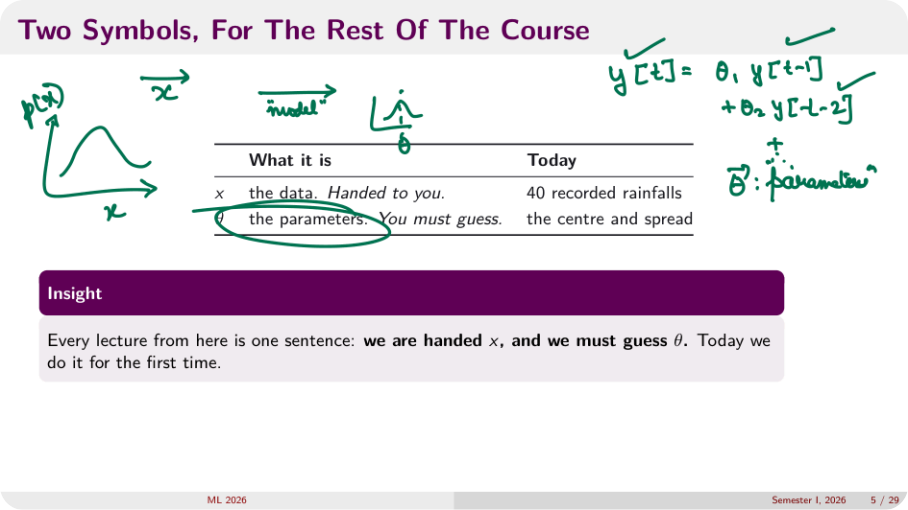
From L2:

- learning = four choices: **data, model, objective, algorithm**;
- $\arg\min$ and convexity;
- types and dimensions as a correctness check.

Careful. We named a **model** but never said where its numbers come from. Today: they are **estimated**, and here is how.

Slide 4 Section: The Estimation Problem

Slide 5 Two Symbols, For The Rest Of The Course



Slide 5 — x is handed to you, θ you must guess (teacher's ink: E[x] = θ₁).

	What it is	Today
x	the data. **Handed to you.**	40 recorded rainfalls
θ	the parameters. **You must guess.**	the centre and spread

Insight: every lecture from here is one sentence: *we are handed x, and we must guess θ.* (Linear regression: $\theta = (w, b)$. Logistic regression: the same. Neural nets: millions of θ's.)

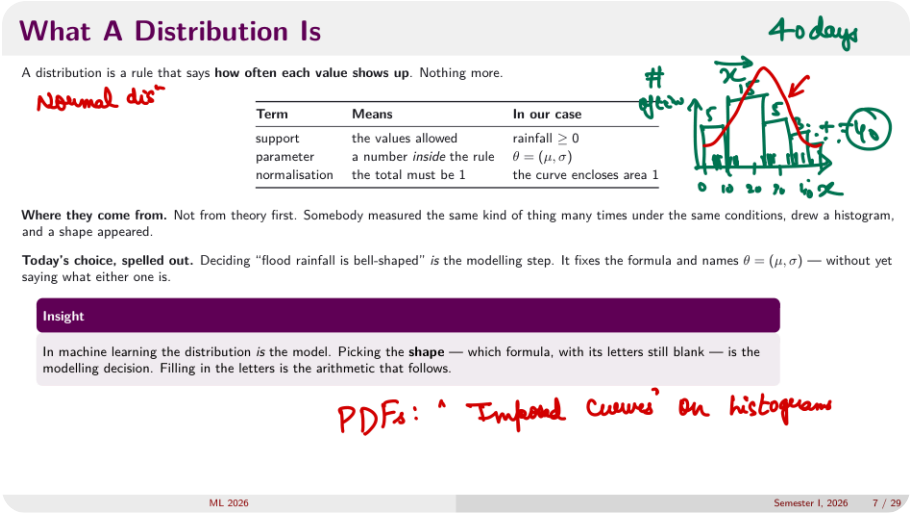
Slide 6 The Setup: Data Comes From Somewhere

$$x_1, x_2, \ldots, x_n \text{ drawn independently from } p(x \mid \theta).$$

We know	We don't know	We want
the **shape** of p (e.g. bell)	the **numbers** θ in it	a good guess $\hat{\theta}$

Careful. Two real assumptions: (1) observations are **independent**; (2) the **shape** we chose is right. Neither is free, and slide 28 comes back to both.

Slide 7 What A Distribution Is



Slide 7 — Board note at the bottom: PDFs are curves imposed on histograms.

A rule that says **how often each value shows up**. Nothing more.

Term	Means	In our case
support	the values allowed	rainfall ≥ 0
parameter	a number *inside* the rule	$\theta = (\mu, \sigma)$
normalisation	the total must be 1	the curve encloses area 1

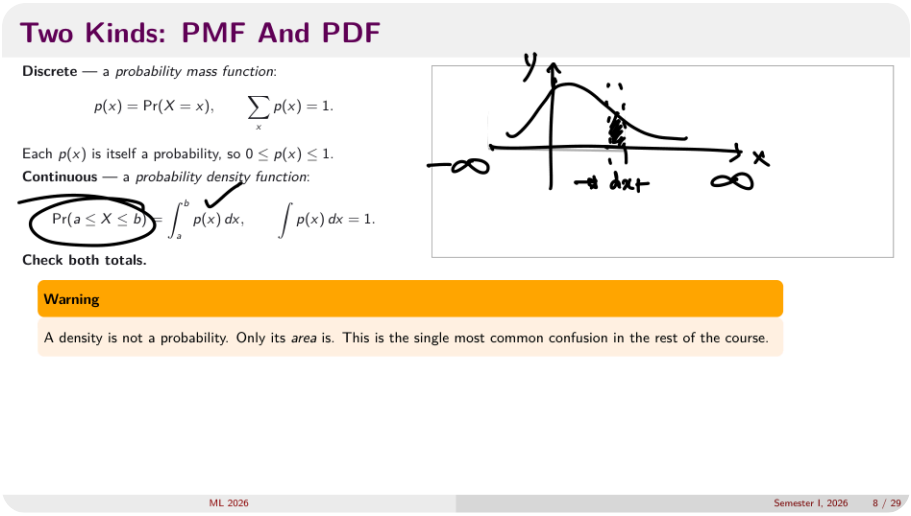
Where they come from. Not from theory first. Someone measured the same kind of thing many times under the same conditions, drew a histogram, and a shape appeared. (Board: PDFs are curves *imposed on histograms*. Take the 40 flood days, histogram them, and lay a bell over the bars.)

Today's choice, spelled out: saying "flood rainfall is bell-shaped" *is* the modelling step. It fixes the formula and names $\theta = (\mu, \sigma)$ without yet saying what either one is.

> Insight: in ML **the distribution is the model**. Choosing the *shape* (a formula with blanks) is the modelling decision. Filling in the blanks is the arithmetic that follows.

(Small catch: a normal has support $\mathbb{R}$, so it puts a little probability on negative rainfall. Around 120 with spread 30, 0 mm is 4σ away, so that probability is negligible. It's still an example of "the shape is only approximately right".)

Slide 8 Two Kinds: PMF And PDF



Slide 8 — Teacher's sketch on the right: probability is the area between a and b.

Discrete (PMF): $p(x) = \Pr(X = x)$, $\sum_x p(x) = 1$, and each $p(x) \in [0, 1]$.

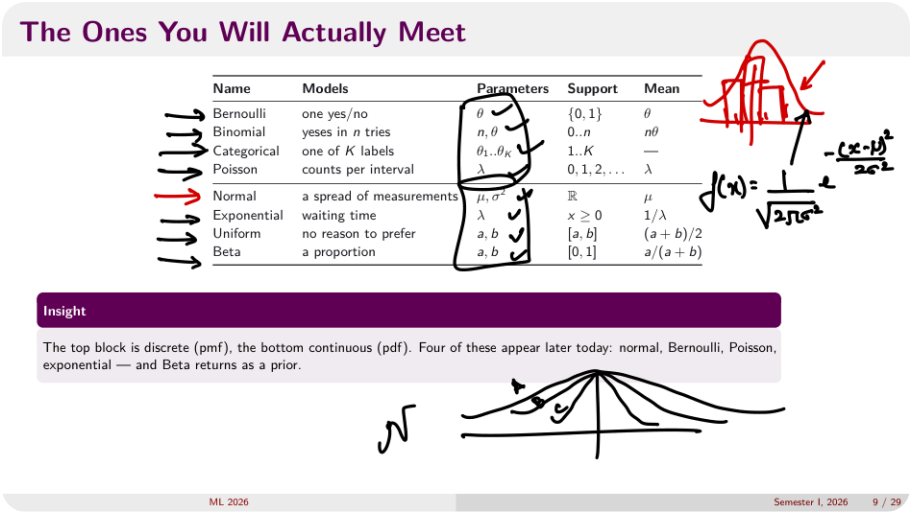
Continuous (PDF): $\Pr(a \le X \le b) = \int_a^b p(x)\,dx$, $\int p(x)\,dx = 1$.

Check both totals (worked):

- Bernoulli: $\sum_{x \in \{0,1\}} p(x) = (1 - \theta) + \theta = 1$ ✓.
- Exponential: $\int_0^\infty \lambda e^{-\lambda x} dx = \left[-e^{-\lambda x} \right]_0^\infty = 0 - (-1) = 1$ ✓.

Careful. **A density is not a probability. Only its area is.** This is the most common confusion in the rest of the course. (See A2: densities can exceed 1.)

Slide 9 The Ones You Will Actually Meet



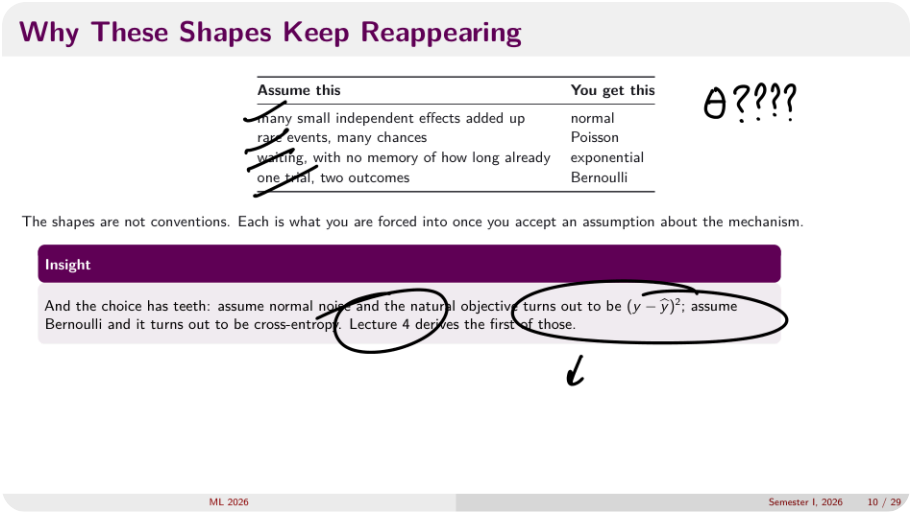
Slide 9 — The ticked rows are the ones used today: normal and Bernoulli.

Name	Models	Parameters	Support	Mean
Bernoulli	one yes/no	θ	$\{0,1\}$	θ
Binomial	yeses in n tries	n,θ	$0..n$	$n\theta$
Categorical	one of K labels	$\theta_1..\theta_K$	$1..K$	—
Poisson	counts per interval	λ	$0,1,2,\ldots$	λ
Normal	a spread of measurements	μ,σ^2	$\mathbb{R}$	μ
Exponential	waiting time	λ	$x \geq 0$	$1/\lambda$
Uniform	no reason to prefer	a,b	$[a,b]$	$(a+b)/2$
Beta	a proportion	a,b	$[0,1]$	$a/(a+b)$

The top four are discrete (PMF); the bottom four are continuous (PDF). Four show up later today: **normal, Bernoulli, Poisson, exponential**. **Beta** comes back as a prior. Full formulas: A3.

Categorical has no "mean" because labels like {cat, dog, bird} aren't numbers you can average.

Slide 10 Why These Shapes Keep Reappearing



Slide 10 — Each shape is the consequence of an assumption about how the data arose.

Assume this	You get this
many small independent effects added up	normal
rare events, many chances	Poisson
waiting, with no memory of how long already	exponential
one trial, two outcomes	Bernoulli

The shapes aren't conventions. Each is forced on you once you accept an assumption about the mechanism (A4).

> **Insight:** the choice has consequences. Normal noise makes the natural objective $(y - \hat{y})^2$. Bernoulli makes it **cross-entropy**. Lecture 4 derives the first.

Slide 11 Today's Question

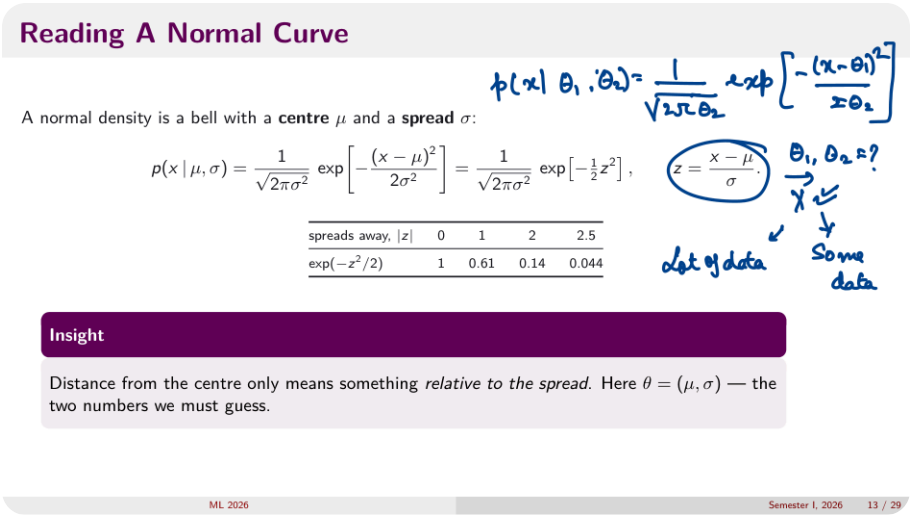
Data: the 40 flood days. The rainfall recorded on each scatters around 120 mm.

Question: where is the curve **centred**, and how **wide** is it? Nobody tells us; we have only the 40 numbers.

Insight: making a bad guess is easy. We need a **rule** that says which guess is best and can defend the answer. That rule is maximum likelihood.

Slide 12 Section: Maximum Likelihood For The Normal

Slide 13 Reading A Normal Curve



Slide 13 — Teacher's ink: z = (x − μ)/σ counts spreads from the centre.

$$p(x \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2} \right] = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\tfrac{1}{2}z^2 \right], \qquad z = \frac{x-\mu}{\sigma}.$$

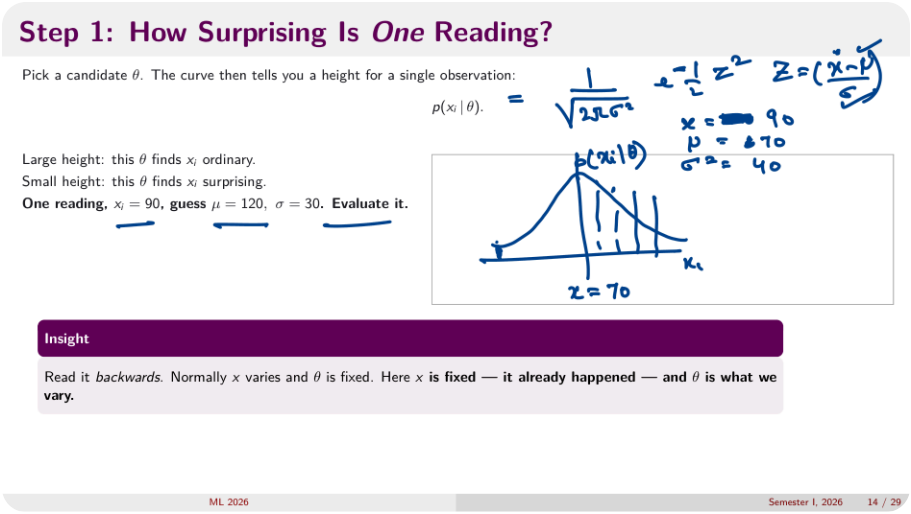
spreads away, $\lvert z \rvert$	0	1	2	2.5
$\exp(-z^2/2)$	1	0.61	0.14	0.044

($e^{-0.5} = 0.607$, $e^{-2} = 0.135$, $e^{-3.125} = 0.044$.)

- μ = **centre** (where the peak is), σ = **spread** (how wide).
- z counts "how many spreads away". The front factor $\frac{1}{\sqrt{2\pi\sigma^2}}$ makes the area 1. A wider bell is also a lower bell.

Insight: distance from the centre only means something **relative to the spread**. 30 mm off is ordinary if $\sigma = 30$ and shocking if $\sigma = 5$. Here $\theta = (\mu, \sigma)$, the two numbers we must guess.

Slide 14 Step 1: How Surprising Is *One* Reading? (solved)



Slide 14 — The sketch: a bell centred at μ, reading x = 90 marked; its height is the score.

Choose a candidate θ. The curve gives a height $p(x_i \mid \theta)$ for one observation. **Large height** means this θ finds x_i ordinary. **Small height** means this θ finds it surprising.

Evaluate: $x_i = 90$, guess $\mu = 120$, $\sigma = 30$.

$$z = \frac{90 - 120}{30} = -1, \qquad \exp(-\tfrac{1}{2} \cdot 1) = 0.6065$$

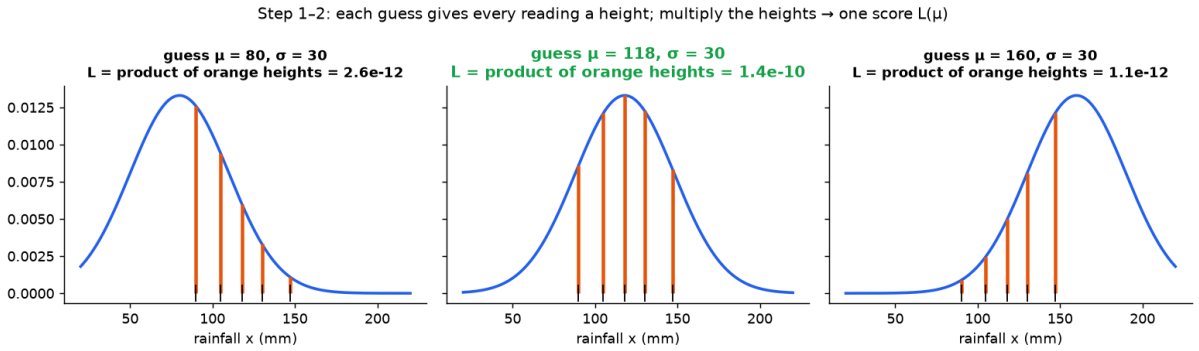
$$\frac{1}{\sqrt{2\pi \cdot 900}} = \frac{1}{30\sqrt{2\pi}} = \frac{1}{75.20} = 0.01330$$

$$p(90 \mid 120, 30) = 0.01330 \times 0.6065 \approx \mathbf{0.0081} \text{ per mm.}$$

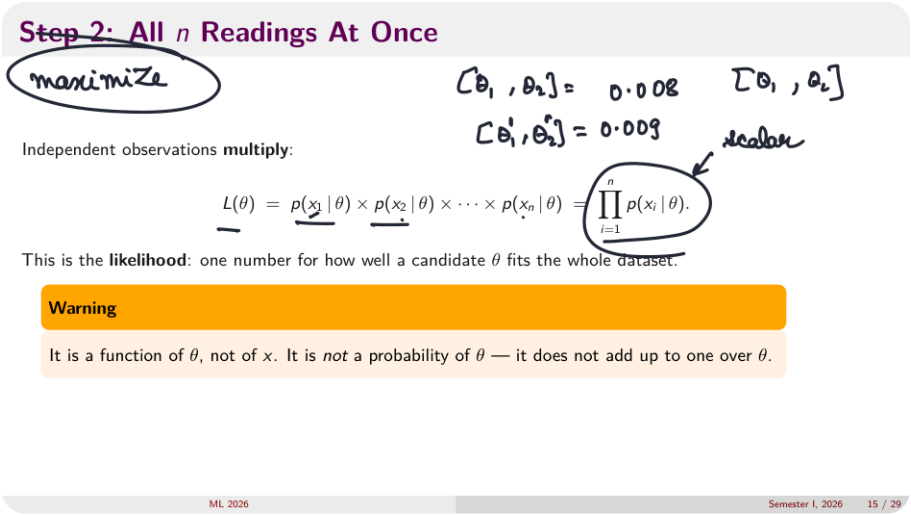
(The same 0.00807 as the flood-likelihood in L2 slide 19.) Compare another guess, $\mu = 90$, $\sigma = 30$: $z = 0$ gives 0.0133, which is higher, so $\mu = 90$ "explains" this one reading better. One reading alone would pull $\hat{\mu}$ onto itself; with all 40, they negotiate.

> **Insight:** read the formula **backwards**. Normally x varies and θ is fixed. Here x is fixed (it already happened) and θ varies.

Slide 15 Step 2: All n Readings At Once



Steps 1–2: slide the bell to a guess μ, read each reading's height, multiply. The middle guess scores highest.



Slide 15 — Two candidate pairs scored 0.008 vs 0.009; maximise, so the second wins.

Independent observations **multiply**:

$$L(\theta) = p(x_1 \mid \theta) \times p(x_2 \mid \theta) \times \cdots \times p(x_n \mid \theta) = \prod_{i=1}^{n} p(x_i \mid \theta).$$

This is the **likelihood**: one number (a scalar) for how well a candidate θ fits the whole dataset. Board example: three readings, each with density around 0.01–0.02, give $L \approx 0.009 \times 0.013 \times 0.02 \approx 2 \times 10^{-6}$. It's already tiny with $n = 3$.

The **multiplication is where independence enters** (L1 §8.6: $P(A \cap B) = P(A)P(B)$ only if independent).

Careful. It is a function of θ, **not** of x. It is **not** a probability of θ: it doesn't sum to one over θ (A5).

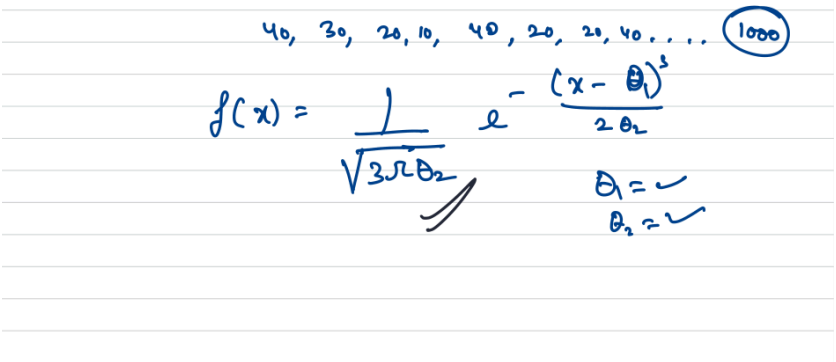
Board, on this slide: two candidates compared by their likelihood score: $L(\theta_1, \theta_2) = 0.008$ vs $L(\theta_1', \theta_2') = 0.009$. The second pair wins, because the job is to **maximise** L. The circled $\prod$ was labelled "scalar": the whole dataset collapses to one number per candidate.

Board pages — What was written in class (annotated PDF pp. 15–16, 18–22)

The annotated deck (`L03b_…`) has seven handwritten pages around slides 14–15. They redo the whole MLE recipe on a tiny dataset, step by step. Notation on the board: θ_1 = centre (μ), θ_2 = spread.

Careful. **Two board slips to watch for.** (1) The board writes $\sqrt{3\pi\theta_2}$ in places; it should be $\sqrt{2\pi\theta_2^2}$ (the normal's front factor is always 2π). (2) θ_2 is used both as the variance (in $2\theta_2$) and as the standard deviation (in $2\theta_2^2$). Pick one: these notes use $\theta_2 = \sigma$, so the exponent is $-(x-\theta_1)^2/2\theta_2^2$.

Board p. 15 — The data and the model



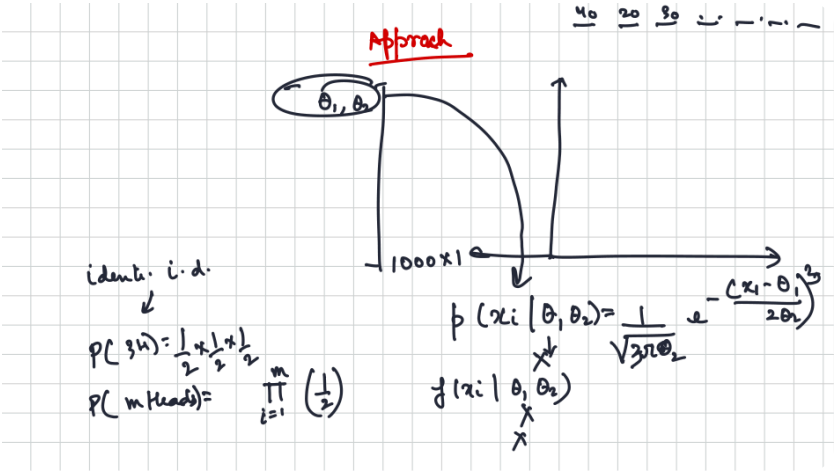
Board p. 15 — Rainfall readings and the normal curve with two unknowns θ_1, θ_2.

Rainfall readings $40, 30, 20, 10, 40, 20, 20, 40, \ldots$ ($n = 1000$ of them). Assume each one comes from a normal curve

$$f(x) = \frac{1}{\sqrt{2\pi\theta_2^2}}\, e^{-(x-\theta_1)^2/2\theta_2^2}.$$

Goal: find the two numbers θ_1, θ_2 (the ticks mean "these are what we'll solve for").

Board p. 16 — The approach, and why we multiply



Board p. 16 — Knobs (θ_1, θ_2) pick a curve; the coin example shows why i.i.d. probabilities multiply.

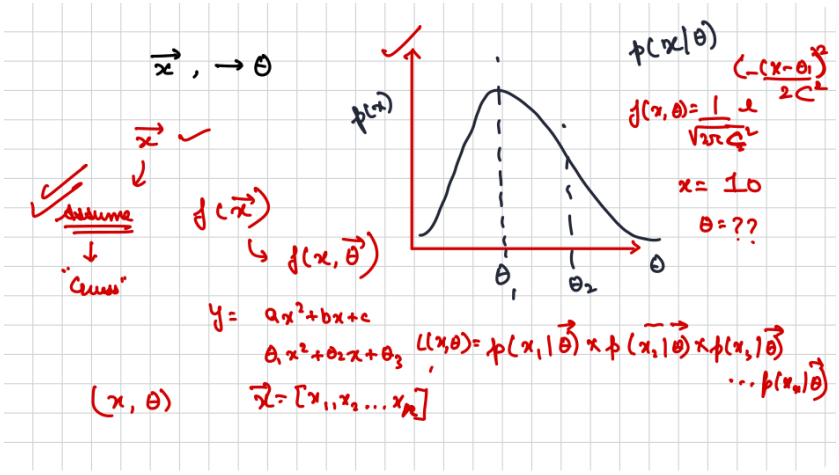
The picture: the knobs (θ_1, θ_2) choose one bell curve; the 1000×1 data vector is then scored by that curve. Each reading gets a height $p(x_i \mid \theta_1, \theta_2)$.

Why multiply? A coin warm-up. Toss a fair coin. The tosses are **i.i.d.** (independent and identically distributed: no toss affects another, and every toss uses the same coin). Then

$$P(3 \text{ heads}) = \tfrac{1}{2} \times \tfrac{1}{2} \times \tfrac{1}{2} = \tfrac{1}{8}, \qquad P(m \text{ heads in a row}) = \prod_{i=1}^{m} \tfrac{1}{2} = \left(\tfrac{1}{2}\right)^m.$$

Same idea for rainfall: independent readings, so their densities multiply.

Board p. 18 — From one reading to a vector of readings

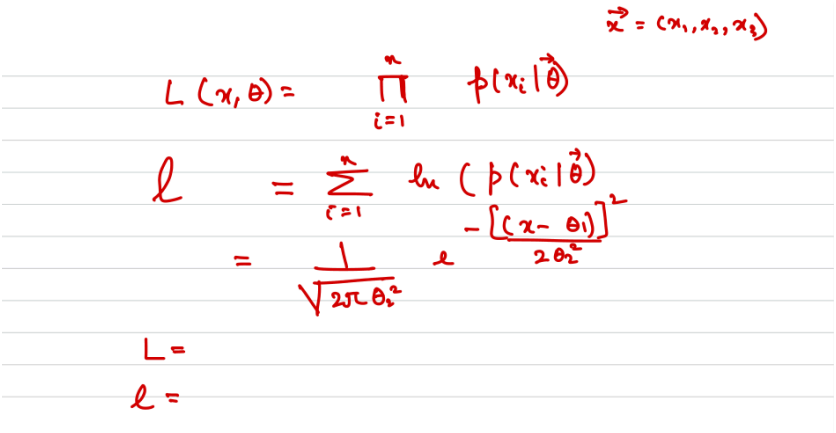


Board p. 18 — Two candidate centres θ_1, θ_2 on one bell; L is a product over all readings.

- Write the data as one vector $\vec{x} = [x_1, x_2, \ldots, x_n]$ and the knobs as $\vec{\theta}$.
- **Assume** a curve shape (the board stresses: this is a *guess*). Writing $f(\vec{x})$ becomes $f(\vec{x}, \vec{\theta})$: the curve depends on its knobs.
- "Knobs" are not special to probability: a parabola $y = ax^2 + bx + c$ is the family $\theta_1 x^2 + \theta_2 x + \theta_3$.
- The sketch: one bell, two candidate centres θ_1 and θ_2 marked on the axis. Different knobs, different heights for the same data.
- Example: one reading $x = 10$, $\theta = ??$ That's the question.

$$L(\vec{x}, \vec{\theta}) = p(x_1 \mid \vec{\theta}) \times p(x_2 \mid \vec{\theta}) \times \cdots \times p(x_n \mid \vec{\theta}).$$

Board p. 19 — Product → sum

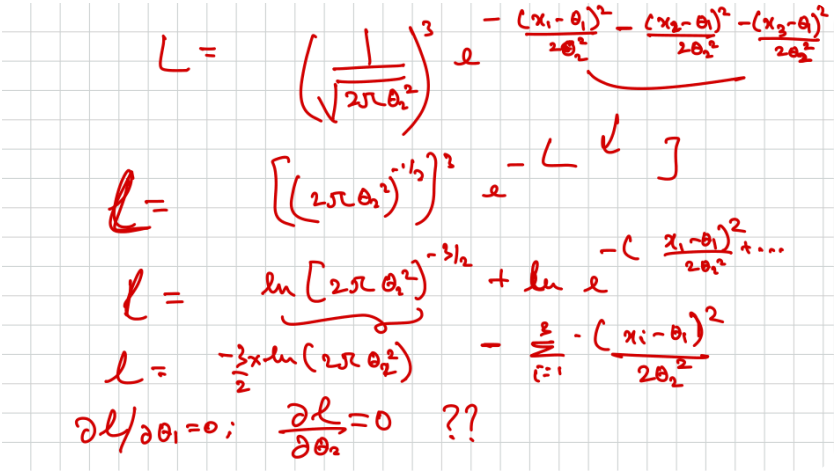


Board p. 19 — Product of densities → sum of logs.

Three readings $\vec{x} = (x_1, x_2, x_3)$:

$$L = \prod_{i=1}^{n} p(x_i \mid \vec{\theta}), \qquad \ell = \ln L = \sum_{i=1}^{n} \ln p(x_i \mid \vec{\theta}).$$

Board p. 20 — Expanding for $n = 3$, step by step



Board p. 20 — n = 3 expanded: exponents add, then take ln.

1. Multiply three bells. Same front factor three times, and exponents add ($e^a e^b = e^{a+b}$):

$$L = \left(\frac{1}{\sqrt{2\pi\theta_2^2}} \right)^3 e^{-\frac{(x_1-\theta_1)^2}{2\theta_2^2} - \frac{(x_2-\theta_1)^2}{2\theta_2^2} - \frac{(x_3-\theta_1)^2}{2\theta_2^2}}.$$

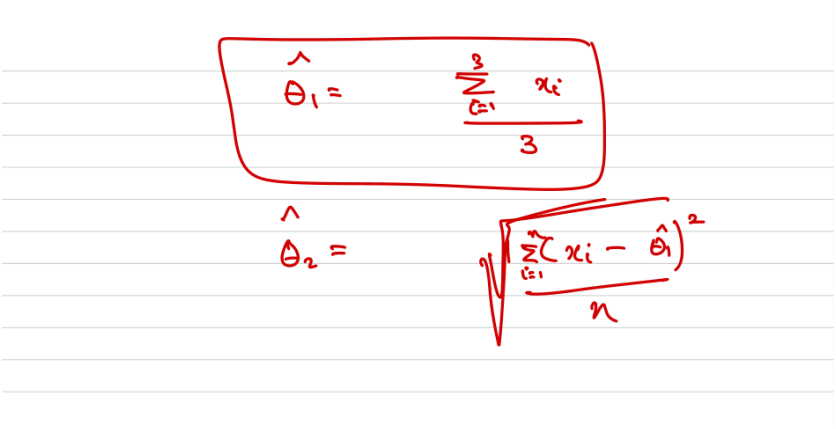
2. Rewrite the front: $\left((2\pi\theta_2^2)^{-1/2}\right)^3 = (2\pi\theta_2^2)^{-3/2}$.

3. Take $\ln$. The log of a product is a sum, and $\ln e^u = u$:

$$\ell = -\tfrac{3}{2}\ln(2\pi\theta_2^2) - \sum_{i=1}^{3}\frac{(x_i-\theta_1)^2}{2\theta_2^2}.$$

4. Two knobs, so two equations: $\dfrac{\partial \ell}{\partial \theta_1} = 0$ and $\dfrac{\partial \ell}{\partial \theta_2} = 0$.

Board p. 21 — The answers

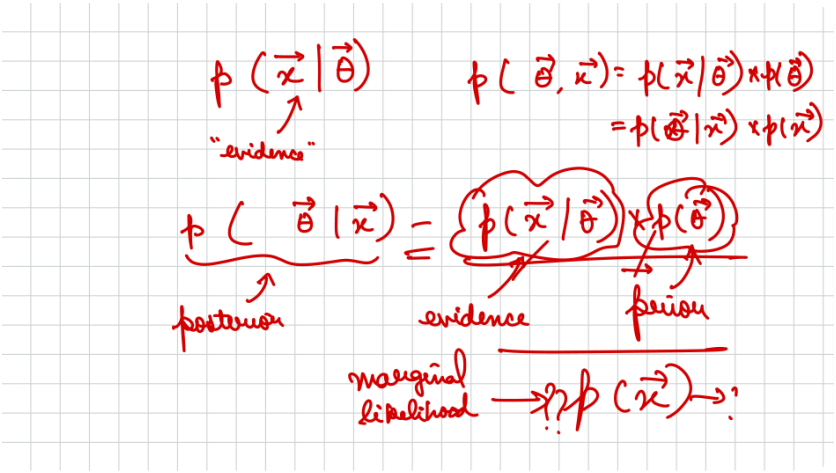


Board p. 21 — Boxed answer: $\hat{\theta}_1$ = average; $\hat{\theta}_2$ = RMS distance from it.

$$\hat{\theta}_1 = \frac{1}{3}\sum_{i=1}^{3} x_i \quad \text{(the average)}, \qquad \hat{\theta}_2 = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i-\hat{\theta}_1)^2} \quad \text{(the RMS distance from the average)}.$$

These are exactly slides 17 and 18 with $n = 3$. Note the $1/n$, not $1/(n-1)$ (slide 18's warning).

Board p. 22 — Bayes' rule for parameters, from the joint



Board p. 22 — Joint split two ways gives posterior = likelihood × prior / evidence.

Start from the joint probability of data and knobs, and split it both ways (L1 §8.4: $P(A \cap B) = P(A \mid B)P(B)$):

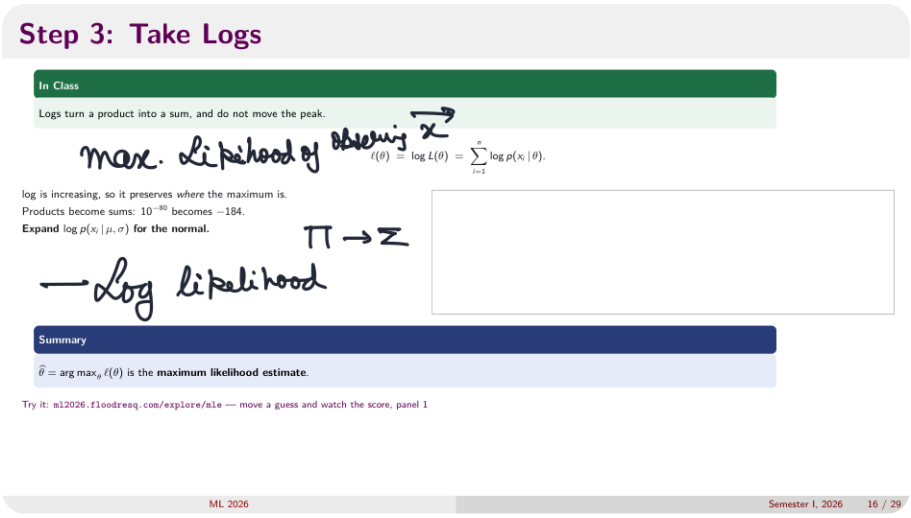
$$p(\vec{\theta}, \vec{x}) = p(\vec{x} \mid \vec{\theta})\, p(\vec{\theta}) = p(\vec{\theta} \mid \vec{x})\, p(\vec{x}).$$

Divide by $p(\vec{x})$:

$$\underbrace{p(\vec{\theta} \mid \vec{x})}_{\text{posterior}} = \frac{\overbrace{p(\vec{x} \mid \vec{\theta})}^{\text{likelihood}}\ \overbrace{p(\vec{\theta})}^{\text{prior}}}{\underbrace{p(\vec{x})}_{\text{evidence / marginal likelihood}}}.$$

- MLE maximises only the likelihood (the cloud the board drew around $p(\vec{x} \mid \vec{\theta})$). Adding the prior and maximising the top line gives **MAP** (slides 25–27).
- **Naming trap.** The board labels $p(\vec{x} \mid \vec{\theta})$ "evidence". In standard usage that term is the **likelihood**; "evidence" means the bottom, $p(\vec{x})$, also called the **marginal likelihood** (also written on the board). It doesn't depend on $\vec{\theta}$, so it never moves the peak.

Slide 16 Step 3: Take Logs (solved)



Slide 16 — Ink: Π → Σ, 'log likelihood'.

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log p(x_i \mid \theta).$$

- $\log$ is increasing, so it keeps the **location** of the maximum (A6).
- Products become sums: 10^{-80} becomes -184 ($= -80 \times \ln 10 = -80 \times 2.3026$).

In Class — expand $\log p(x_i \mid \mu, \sigma)$ for the normal. Use $\log(ab) = \log a + \log b$ and $\log e^u = u$:

$$\log p(x_i \mid \mu, \sigma) = \log \frac{1}{\sqrt{2\pi\sigma^2}} + \log \exp\Big[-\frac{(x_i - \mu)^2}{2\sigma^2} \Big] = -\tfrac{1}{2} \log(2\pi\sigma^2) - \frac{(x_i - \mu)^2}{2\sigma^2}.$$

Sum over the n readings (the first term is the same every time, so it's multiplied by n):

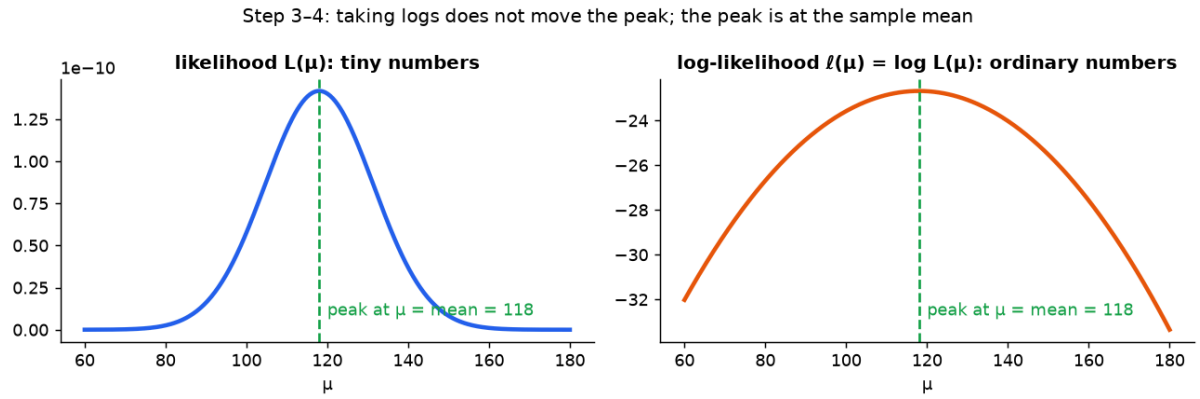
$$\boxed{\ell(\mu, \sigma) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2.}$$

The second term is the negative **sum of squared errors** around μ, scaled by $1/2\sigma^2$. Squared error is already appearing.

Summary: $\hat{\theta} = \arg\max_\theta \ell(\theta)$ is the **maximum likelihood estimate (MLE)**.

Try it: Open the interactive sections. Move a guess and watch the score.

Slide 17 Step 4: Find The Peak — The Centre (solved)



Steps 3–4: L and log L peak at the same μ, the sample mean. Logs just make the numbers readable.

With σ fixed:

$$\ell(\mu) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

Only the second term contains μ.

Differentiate (chain rule: $\frac{d}{d\mu}(x_i-\mu)^2 = 2(x_i-\mu)\cdot(-1)$):

$$\frac{d\ell}{d\mu} = -\frac{1}{2\sigma^2}\sum_{i} 2(x_i-\mu)(-1) = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu).$$

Set to zero and solve:

$$\sum_{i}(x_i-\mu) = 0 \;\Rightarrow\; \sum_{i} x_i - n\mu = 0 \;\Rightarrow\; \boxed{\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}.}$$

Check it's a max: $\frac{d^2\ell}{d\mu^2} = -\frac{n}{\sigma^2} < 0$ ✓ (a downward parabola in μ, so this is the only peak).

Notice: maximising $\ell(\mu)$ is the same as **minimising** $\sum(x_i-\mu)^2$. MLE under a normal = least squares. And σ dropped out, so $\hat{\mu}$ is the same whatever σ is.

Summary: the MLE of the centre is the **sample mean**.

Slide 18 Step 4 Again, For The Spread (solved)

Write $v = \sigma^2$ (differentiating with respect to the variance is cleaner) and plug in $\hat{\mu}$:

$$\ell(v) = -\frac{n}{2}\log(2\pi v) - \frac{1}{2v}\sum_i (x_i - \hat{\mu})^2.$$

Let $S = \sum_i (x_i - \hat{\mu})^2$ (a fixed number once the data are in). Note $\log(2\pi v) = \log 2\pi + \log v$.

Differentiate:

$$\frac{d\ell}{dv} = -\frac{n}{2}\cdot\frac{1}{v} - S\cdot\frac{d}{dv}\left(\frac{1}{2v}\right) = -\frac{n}{2v} + \frac{S}{2v^2}.$$

Set to zero: multiply by $2v^2$, giving $-nv + S = 0$:

$$\boxed{\hat{v} = \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{\mu})^2, \qquad \hat{\sigma} = \sqrt{\hat{\sigma}^2}.}$$

Check: for $v < S/n$ the derivative is positive, and for $v > S/n$ it's negative. So the function rises then falls, and this is a max ✓.

Careful. Note the $1/n$, **not** $1/(n-1)$. This estimate of the spread is slightly **too small** on average. Reason: $\hat{\mu}$ was fitted to the same data, so the data sit closer to $\hat{\mu}$ than to the true μ. In fact $\mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n}\sigma^2$. Dividing by $n-1$ instead fixes this ("Bessel's correction"). For $n = 40$ the factor is 0.975, which is tiny. L8 explains why (bias of estimators).

Slide 19 On Our Forty Days

$$\text{centre: } \hat{\mu} = \frac{1}{n}\sum x_i, \qquad \text{spread: } \hat{\sigma} = \sqrt{\frac{1}{n}\sum (x_i - \hat{\mu})^2}.$$

Add the forty readings and divide by forty. Then do the same for the squared deviations and take the square root.

> **Insight:** the forty days give $\hat{\mu} \approx 118$ mm and $\hat{\sigma} \approx 31$ mm. Rounded, that's the "centred 120, spread 30" used in L2's flood example. **Those numbers were never given to us**; they were estimated from recorded days exactly like this. Real estimates don't come out round.

Try it: Open the interactive sections, the same forty points.

Slide 20 Section: The Method Beyond The Normal

Slide 21 None Of The Steps Used The Normal

The four steps:

1. write $p(x_i \mid \theta)$ for one observation;
2. multiply over the n observations;
3. take logs;
4. differentiate, set to zero, solve for θ.

Swap the distribution and **only line 1 changes**:

Data looks like	Use	MLE turns out to be
a spread of measurements	normal	the sample mean
yes/no outcomes	Bernoulli	the fraction of yeses, k/n
counts per interval	Poisson	the average count
waiting times	exponential	$1/$the average wait

> **Worked: Poisson.** $p(x_i \mid \lambda) = \lambda^{x_i} e^{-\lambda} / x_i!$

$$\ell(\lambda) = \sum_i \left[x_i \log \lambda - \lambda - \log x_i! \right] = \left(\sum_i x_i \right) \log \lambda - n\lambda - \text{const}$$

$$\ell'(\lambda) = \frac{\sum x_i}{\lambda} - n = 0 \;\Rightarrow\; \hat{\lambda} = \frac{1}{n}\sum x_i = \bar{x}.$$

Worked: Exponential. $p(x_i \mid \lambda) = \lambda e^{-\lambda x_i}$

$$\ell(\lambda) = n\log\lambda - \lambda\sum_i x_i, \qquad \ell'(\lambda) = \frac{n}{\lambda} - \sum_i x_i = 0 \;\Rightarrow\; \hat{\lambda} = \frac{n}{\sum x_i} = \frac{1}{\bar{x}}.$$

This makes sense: if the average wait is 5 days, events happen at a rate of 1/5 per day.

Slide 22 Example Two: Steps 1 To 3, For A Bernoulli (solved)

A concrete cube either **fails** its strength test or **passes**. In n tests, k failed. θ = Pr(fail).

Step 1. One observation: it scores θ if it failed and $1-\theta$ if it passed. With $x_i = 1$ for fail:

$$p(x_i \mid \theta) = \theta^{x_i}(1-\theta)^{1-x_i}.$$

Step 2. Multiply over the n tests: each failure contributes a θ and each pass a $(1-\theta)$, so

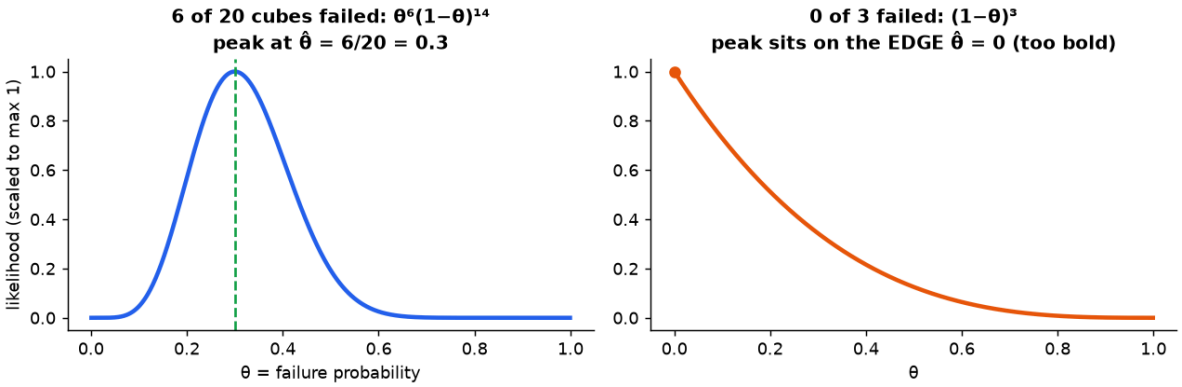
$$L(\theta) = \underbrace{\theta\cdots\theta}_{k}\;\underbrace{(1-\theta)\cdots(1-\theta)}_{n-k} = \theta^k(1-\theta)^{n-k}.$$

Step 3. Take logs:

$$\ell(\theta) = k\log\theta + (n-k)\log(1-\theta).$$

Insight: the same three moves, and the powers collect themselves. The order of passes and fails doesn't matter; only the **count** k does.

Slide 23 Example Two: Step 4, And The Answer (solved)



Bernoulli likelihoods: 6 of 20 peaks at 0.3; 0 of 3 peaks on the edge at 0 (the problem slide 25 fixes).

Differentiate (A6):

$$\ell'(\theta) = \frac{k}{\theta} - \frac{n-k}{1-\theta}.$$

Set to zero, multiply out, collect the θ terms:

$$\frac{k}{\theta} = \frac{n-k}{1-\theta} \;\Rightarrow\; k(1-\theta) = (n-k)\theta \;\Rightarrow\; k - k\theta = n\theta - k\theta \;\Rightarrow\; \boxed{\hat{\theta} = \frac{k}{n}.}$$

Check max: $\ell''(\theta) = -\frac{k}{\theta^2} - \frac{n-k}{(1-\theta)^2} < 0$ ✓ (concave).

> **Summary:** 6 of 20 cubes failed, so $\hat{\theta} = 6/20 = 0.3$. Two examples, one method: write $p(x \mid \theta)$, multiply, take logs, find the peak.

Try it: Open the interactive sections, the same 6 of 20 and two more distributions.

Slide 24

Section: Priors, MAP, And Uncertainty

Slide 25

When *k/n* Is Not Usable

A small survey: three cubes tested, none failed. So $n = 3$, $k = 0$, and MLE says $\hat{\theta} = 0/3 = 0$.

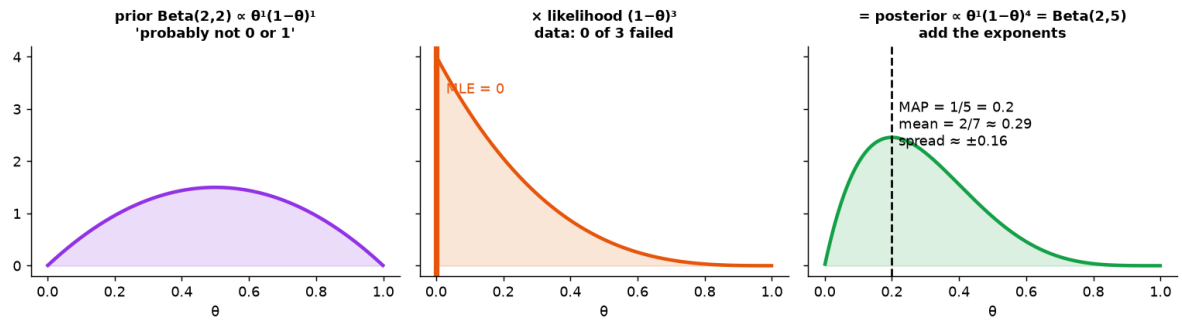
Careful. That answer says failure **never** happens. Three tests are not enough to support it.

Look at the likelihood: $L(\theta) = (1-\theta)^3$ is largest at $\theta = 0$, the **edge** of $[0, 1]$. Setting the derivative to zero doesn't even apply here: $\ell'(\theta) = -3/(1-\theta)$ is never 0. The max is at the boundary (A7 step 4).

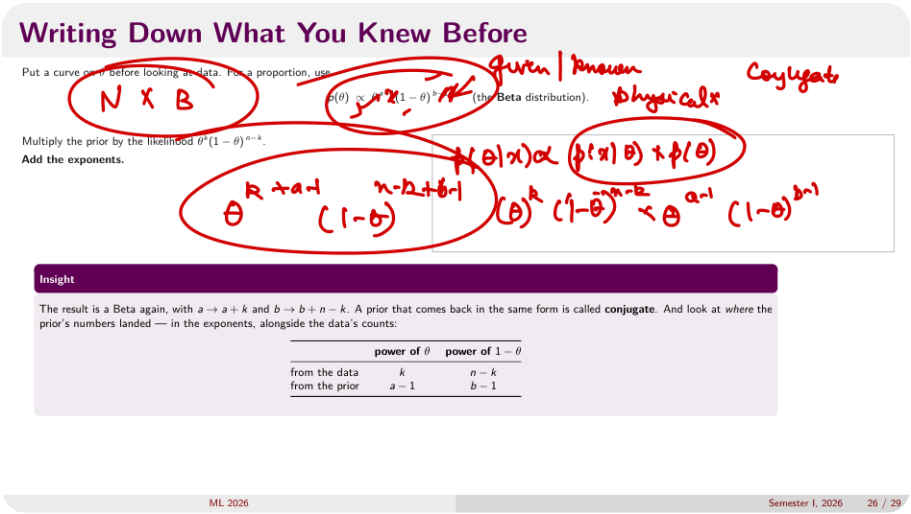
Insight: MLE uses the data **and nothing else**. With little data the peak can sit on the edge, and that edge is a claim we wouldn't make. (The same failure appears in language models and Naive Bayes as the "zero-count problem", fixed by add-one/Laplace smoothing, which is exactly the MAP result below.)

Slide 26

Writing Down What You Knew Before (solved)



Prior × likelihood = posterior, drawn: Beta(2,2) × (1–θ)³ = Beta(2,5). MAP = 0.2 instead of the MLE's 0.



Slide 26 — Ink: posterior ∝ likelihood × prior; add the exponents → conjugate.

Put a curve on θ **before** looking at data. For a proportion, use the **Beta** distribution:

$$p(\theta) \propto \theta^{a-1}(1-\theta)^{b-1}.$$

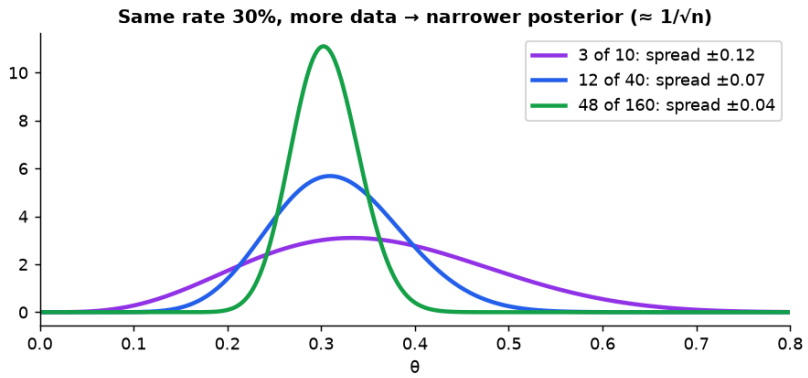
Multiply the prior by the likelihood $\theta^k(1-\theta)^{n-k}$ (A8: posterior ∝ likelihood × prior), then **add the exponents**:

$$p(\theta \mid x) \propto \theta^k(1-\theta)^{n-k} \cdot \theta^{a-1}(1-\theta)^{b-1} = \theta^{a+k-1}(1-\theta)^{b+n-k-1}.$$

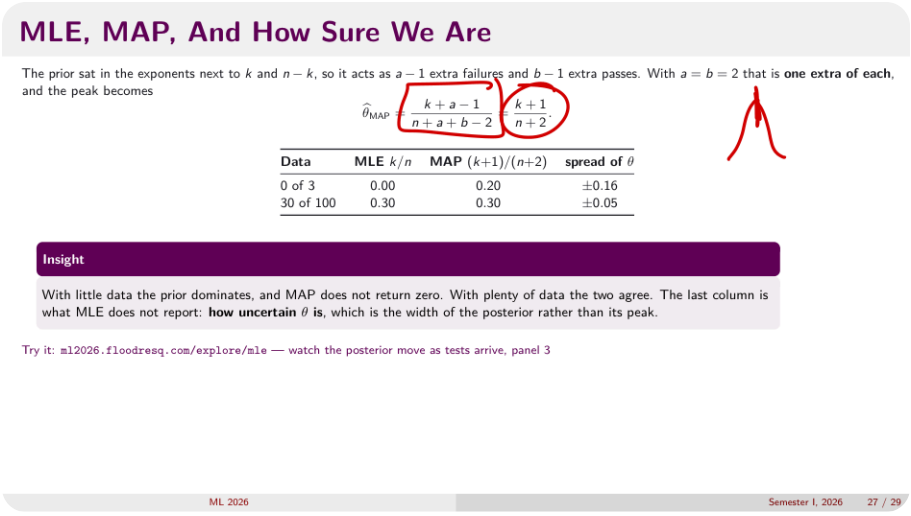
> **Insight:** the result is a Beta again, with $a \to a+k$ and $b \to b+n-k$. A prior that comes back in the same form is called **conjugate** (A9). Look at **where** the prior's numbers landed: in the exponents, next to the data's counts:

	power of θ	power of $1-\theta$
from the data	k	$n-k$
from the prior	$a-1$	$b-1$

Slide 27 MLE, MAP, And How Sure We Are (solved)



Same 30% rate, more tests: the posterior gets narrower, so we become surer.



Slide 27 — The sketch at the top: the prior is a hump; the table compares MLE, MAP and spread.

The prior sits in the exponents next to k and $n-k$, so it acts as $a-1$ **extra failures and** $b-1$ **extra passes**. Find the peak of the posterior by repeating slide 23 with $k \to k+a-1$ and $n \to n+a+b-2$:

$$\hat{\theta}_{\text{MAP}} = \frac{k+a-1}{n+a+b-2}.$$

With $a=b=2$ (one extra of each):

$$\hat{\theta}_{\text{MAP}} = \frac{k+1}{n+2}.$$

Data	MLE k/n	MAP $(k+1)/(n+2)$	spread of θ
0 of 3	0.00	0.20	±0.16
30 of 100	0.30	0.30	±0.05

Where the last column comes from. The posterior is Beta(α, β) with $\alpha = a + k$, $\beta = b + n - k$. Its standard deviation is

$$\text{spread} = \sqrt{\frac{\bar{\theta}(1-\bar{\theta})}{\alpha+\beta+1}}, \qquad \bar{\theta} = \frac{\alpha}{\alpha+\beta} \text{ (posterior mean)}.$$

That's the familiar $\sqrt{\bar{\theta}(1-\bar{\theta})/n}$ with the prior's counts included.

Verify both rows:

- **0 of 3:** $\alpha = 2$, $\beta = 2 + 3 = 5$. MAP $= 1/5 = 0.20$. $\bar{\theta} = 2/7 = 0.286$. Spread $= \sqrt{0.286 \times 0.714/8} = \sqrt{0.0255} = 0.16$ ✓.
- **30 of 100:** $\alpha = 32$, $\beta = 72$. MAP $= 31/102 = 0.304 \approx 0.30$. $\bar{\theta} = 32/104 = 0.308$. Spread $= \sqrt{0.308 \times 0.692/105} = \sqrt{0.00203} = 0.045 \approx 0.05$ ✓.

> Insight: with **little data the prior dominates**, so MAP doesn't return zero. With **plenty of data** the two agree. The last column is what MLE doesn't report: **how uncertain θ is**, which is the *width* of the posterior rather than its peak. With 0 of 3 we know very little (±0.16). With 100 tests we know a lot more (±0.05). The spread shrinks roughly like $1/\sqrt{n}$.

Try it: Open the interactive sections. Watch the posterior move as tests arrive.

Slide 28

What We Just Assumed

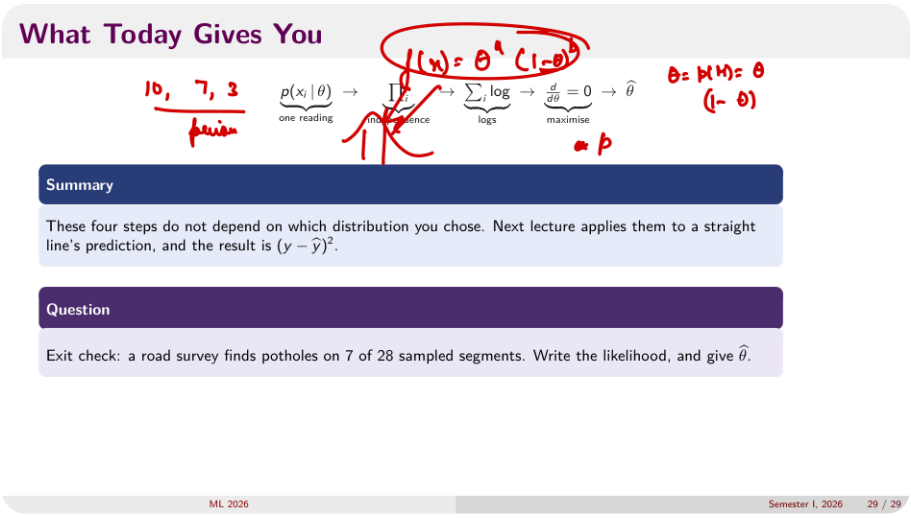
Assumption	Where it entered	When it breaks
independence	the multiplication in Step 2	sensors in a row, days in a storm
the shape is right	writing $p(x \mid \theta)$ at all	a bell fitted to lopsided data
one best answer	setting the derivative to zero	several peaks, or a flat ridge

Careful. Maximum likelihood answers "which θ makes my data **least surprising**". It does **not** say the model is right, and it returns **one number with no uncertainty attached**.

What each failure looks like:

- **Dependence:** 40 correlated storm days carry less information than 40 independent ones, so you'll be over-confident.
- **Wrong shape:** rainfall is often right-skewed (long upper tail). A symmetric bell fitted to it puts its centre and spread in misleading places. The MLE is still "the best bell", just best within the wrong family.
- **Non-unique peak:** e.g. two features that are copies of each other, so infinitely many (w_1, w_2) fit equally well (a flat ridge). Regularisation/MAP (L7) picks one.

Slide 29 What Today Gives You



Slide 29 — Ink: the exit check's likelihood $\theta^k(1-\theta)^{n-k}$.

$$\underbrace{p(x_i \mid \theta)}_{\text{one reading}} \to \underbrace{\prod_i}_{\text{independence}} \to \underbrace{\sum_i \log}_{\text{logs}} \to \underbrace{\frac{d}{d\theta} = 0}_{\text{maximise}} \to \hat{\theta}$$

Summary: these four steps don't depend on which distribution you chose. Next lecture applies them to a **straight line's prediction**, and the result is $(y - \hat{y})^2$.

Exit check (solved): a road survey finds potholes on 7 of 28 sampled segments.

- Likelihood (Bernoulli, $k = 7$, $n = 28$): $L(\theta) = \theta^7(1-\theta)^{21}$.
- Log-likelihood: $\ell(\theta) = 7\log\theta + 21\log(1-\theta)$.
- $\ell'(\theta) = \frac{7}{\theta} - \frac{21}{1-\theta} = 0 \Rightarrow 7(1-\theta) = 21\theta \Rightarrow \hat{\theta} = \frac{7}{28} = \mathbf{0.25}$.
- (Bonus, MAP with Beta(2,2): $\frac{8}{30} \approx 0.267$.)

Also worth knowing

How sharp is the peak? The negative second derivative of the log-likelihood at the peak is the **Fisher information** (observed). For a Bernoulli with n tosses, $I(\hat{\theta}) = n/\big(\hat{\theta}(1-\hat{\theta})\big)$, and the standard error is about $1/\sqrt{I}$. Four times the data halves the standard error.

Averages form a bell. By the central limit theorem, the mean of n independent draws is approximately normal with spread $\sigma/\sqrt{n}$, whatever the shape of the original distribution.

Modelling a skewed positive quantity. X is **lognormal** when $\log X$ is normal. Times and sizes often behave this way. Writing $y_i = \log x_i$, the MLEs are the mean and variance of the y_i. The median of X is e^{θ_1}, and the long right tail pulls the mean up to $e^{\theta_1 + \theta_2/2}$.

A conjugate prior for a variance. With the mean known, the likelihood has the shape $\theta_2^{-n/2} e^{-S/(2\theta_2)}$. The **inverse-gamma** prior $\theta_2^{-a-1} e^{-b/\theta_2}$ has the same shape, so powers and constants simply add: $\mathrm{IG}(a, b)$ becomes $\mathrm{IG}(a + \frac{n}{2},\ b + \frac{S}{2})$. With $a = 3$, $b = 4$, $n = 8$ and $S = 16$, the posterior is $\mathrm{IG}(7, 12)$. Its mode is $b/(a+1) = 1.5$, midway between the prior's 1 and the data's 2, and its mean is $b/(a-1) = 2$.

Part C Practice

1. Is each a valid PMF/PDF? (a) $p(0) = 0.3, p(1) = 0.8$. (b) $p(x) = 3$ on $[0, \frac{1}{3}]$, else 0. (c) $p(x) = 2x$ on $[0, 1]$.
2. $x = 150$ mm, $\mu = 120$, $\sigma = 30$. Compute $p(x \mid \mu, \sigma)$. Is it more or less surprising than $x = 90$?
3. Rainfalls $(100, 120, 140)$. Find $\hat{\mu}$ and $\hat{\sigma}^2$ (MLE). What would the $1/(n-1)$ version give?
4. Write $\ell(\mu)$ for $(100, 120, 140)$ with $\sigma = 20$, and evaluate it at $\mu = 120$ and $\mu = 110$. Which is bigger, and why did you expect that?
5. Why is $\arg\max_\theta L(\theta) = \arg\max_\theta \log L(\theta)$? Would $\arg\max_\theta \sqrt{L(\theta)}$ also be the same? $\arg\max_\theta(-L(\theta))$?
6. Daily count of trucks over a bridge in a 5-minute window: $3, 5, 2, 6, 4$. Assuming Poisson, find $\hat{\lambda}$.
7. Times between pump failures (days): $10, 30, 20$. Assuming exponential, find $\hat{\lambda}$ and interpret it.
8. Derive the MLE of θ for a Binomial: k successes in n tries, $p(k \mid \theta) = \binom{n}{k}\theta^k(1-\theta)^{n-k}$. What happened to $\binom{n}{k}$?
9. 0 of 10 sensors failed. Give MLE, MAP with Beta(2,2), and MAP with Beta(1,1).
0. Prior Beta(3,7), data 4 failures in 10. What's the posterior? Its mean and MAP?
1. A friend says "the likelihood of $\theta = 0.3$ is 0.002, so there's a 0.2% chance $\theta = 0.3$." What's wrong?
2. You fit a normal to 1000 hourly river levels from one month. List two assumptions from slide 28 that are likely violated and what each does to your estimate.
3. Show that MAP = $\arg\max_\theta[\ell(\theta) + \log p(\theta)]$, and say which term "regularises".

1. (a) No: $0.3 + 0.8 = 1.1 \neq 1$. (b) Yes: area $= 3 \times \frac{1}{3} = 1$ (density > 1 is fine). (c) Yes: $\int_0^1 2x\,dx = [x^2]_0^1 = 1$, and $2x \geq 0$.
2. $z = 1$, so the same as $x = 90$: $p \approx 0.0081$. Equally surprising, since the normal is symmetric.
3. $\hat{\mu} = 120$. Deviations $-20, 0, 20$; squares sum to 800. $\hat{\sigma}^2 = 800/3 \approx 266.7$ ($\hat{\sigma} \approx 16.3$). The $1/(n-1)$ version: $800/2 = 400$ ($s = 20$).
4. $\ell(\mu) = -\frac{3}{2}\log(2\pi \cdot 400) - \frac{1}{800}\sum(x_i - \mu)^2$. Constant $= -1.5\ln(2513.3) = -1.5 \times 7.829 = -11.74$. At 120: SSE $= 800$, so $\ell = -11.74 - 1 = -12.74$. At 110: deviations $-10, 10, 30$ give SSE $= 1100$, so $\ell = -11.74 - 1.375 = -13.12$. $\mu = 120$ is bigger, as expected, since it's the sample mean (the MLE).
5. $\log$ is strictly increasing, so it preserves order and therefore the argmax. $\sqrt{\cdot}$ is also strictly increasing on $[0, \infty)$, so yes, the same. $-L$ flips the order: $\arg\max(-L) = \arg\min L$, which is different.

6. $\hat{\lambda} = \bar{x} = 20/5 = 4$ trucks per window.
7. $\bar{x} = 20$ days, so $\hat{\lambda} = 1/20 = 0.05$ failures per day: one failure per 20 days on average.
8. $\ell = \log \binom{n}{k} + k \log \theta + (n-k)\log(1-\theta)$. The first term has no θ, so it vanishes when you differentiate. The rest is slide 23, giving $\hat{\theta} = k/n$. Constants never move the argmax.
9. MLE $= 0/10 = 0$. MAP Beta(2,2) $= 1/12 \approx 0.083$. MAP Beta(1,1) $= 0/10 = 0$ (a flat prior adds no pseudo-counts, so MAP = MLE).
10. Beta$(3+4,\ 7+6)$ = Beta(7, 13). Mean $= 7/20 = 0.35$. MAP $= 6/18 \approx 0.333$. (The MLE alone would be 0.4; the prior, centred at 0.3, pulls it down.)
11. Likelihood is not a probability of θ (it doesn't sum to 1 over θ, and θ is continuous so $\Pr(\theta = 0.3) = 0$ anyway). It's a relative score. To make probability statements about θ you need a prior and the posterior.
12. **Independence:** consecutive hours are strongly correlated, so the effective sample size is far below 1000 and you're over-confident (the spread of θ is too small). **Shape:** river levels are skewed with flood spikes, so a bell under-represents extremes. Also **identical distribution:** one month may not represent other seasons.
13. $p(\theta \mid x) \propto p(x \mid \theta)p(\theta)$. Take logs (increasing, so the argmax is unchanged) and drop the evidence (no θ): $\arg\max[\log p(x \mid \theta) + \log p(\theta)] = \arg\max[\ell(\theta) + \log p(\theta)]$. The $\log p(\theta)$ term regularises: it penalises θ values the prior finds implausible (e.g. $\theta = 0$ under Beta(2,2) has $\log p = -\infty$).